

2.3 Change of Parameter; Arc Length Worksheet

1. Determine if the following functions are smooth.

a) $\mathbf{r}(t) = \left\langle \frac{1}{2}t^2 - t, \frac{1}{3}t^3 + \frac{1}{2}t^2 - 2t, \ln(t) \right\rangle$ where $0 < t < \infty$.

b) $\mathbf{r}(t) = \cos(t+1)\mathbf{i} + (e^{t+1} - t)\mathbf{j} + \mathbf{k}$

c) $\mathbf{r}(t) = \sin(t)\mathbf{i} + \cos(t)\mathbf{j}$

d) $\mathbf{r}(t) = \langle \sqrt{t}, t, t \rangle$ where $0 \leq t \leq 2$.

2. Find the length of the curve given by $\mathbf{r}(t) = \langle 2, 2t^2, t^3 \rangle$ for $1 \leq t \leq 2$.

3. Consider the curve $\mathbf{r}(t) = \langle \cos(t), \sin(t), t \rangle$ for $0 \leq t \leq 6\pi$.

a) Find the length of $\mathbf{r}(t)$.

b) Create an arc length parameterization for the curve starting from when $t = 0$.

c) What point do you reach if travel along the curve $\mathbf{r}(t)$ for $\pi\sqrt{2}$ units, starting from when $t = 0$?

4. Consider the curve $\mathbf{r}(t) = \left\langle t^2, \frac{8}{3}t^{3/2}, 4t \right\rangle$ for $0 \leq t \leq 10$.

a) Find the length of $\mathbf{r}(t)$.

b) Create an arc length function for the curve starting from when $t = 5$.

c) Explain why it would be difficult to create an arc length parameterization for this curve starting from when $t = 5$.

5. Consider the curve $\mathbf{r}(t) = \langle e^{2t}, e^{-2t}, 2\sqrt{2}t \rangle$ for $0 \leq t \leq 5$.

a) Find the length of $\mathbf{r}(t)$.

b) Create an arc length function for the curve with starting from $t = 0$.

c) Reparametrize $\mathbf{r}(t)$ so that it is traced in the opposite direction.

6. Consider the curve $\mathbf{r}(t) = \left\langle \frac{\sqrt{10}}{2}t^2, \frac{1}{3}t^3, 5t + 14 \right\rangle$ for $0 \leq t \leq 2$.

a) Find the length of $\mathbf{r}(t)$.

b) Create an arc length function for the curve with starting from $t = 0$.

c) Reparametrize $\mathbf{r}(t)$ so that it is traced in the opposite direction for $10 \leq \tau \leq 20$

7. Consider the curve $\mathbf{r}(t) = \left\langle 3t, 4t, \frac{2}{3}t^{3/2} \right\rangle$ for $0 \leq t \leq 1$.

a) Find the length of $\mathbf{r}(t)$.

b) Create an arc length parameterization for the curve starting from when $t = 0$.

c) Reparametrize $\mathbf{r}(t)$ so that it is traced in the opposite direction.